Software Engineering Lab Report

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Aim

To study project management techniques using a Gantt chart and plan the development phases of a DeepFake Detection System.

Software & Tools Used

- Microsoft Excel / Google Sheets
- Online Gantt Chart Tools
- Python
- TensorFlow / PyTorch
- OpenCV

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Theory

Project management is essential for successful software development. It involves planning, scheduling, and monitoring project activities. A Gantt chart is a visual tool that represents tasks over time, helping in understanding task dependencies, durations, and progress tracking.

In this experiment, a Gantt chart is used to plan the development of a DeepFake Detection System, including phases such as data collection, preprocessing, model training, evaluation, GUI development, and deployment.

Methodology

1. Identify all project phases
2. Divide the project into smaller tasks
3. Assign timelines to each task
4. Arrange tasks sequentially
5. Create a Gantt chart
6. Review and optimize the schedule

Implementation Procedure

1. Define the project scope
2. Identify key stages of development

3. Break stages into smaller tasks
4. Assign duration and sequence
5. Create Gantt chart using Excel or tools
6. Verify dependencies and overlaps
7. Optimize the schedule

Project Task Scheduling

Task	Start Week	End Week
Requirement Analysis	Week 1	Week 2
Dataset Collection	Week 3	Week 4
Data Preprocessing	Week 5	Week 6
Model Training	Week 7	Week 9
Model Evaluation	Week 10	Week 11
GUI Development	Week 12	Week 13
Testing & Deployment	Week 14	Week 15

Results

Gantt Chart Representation

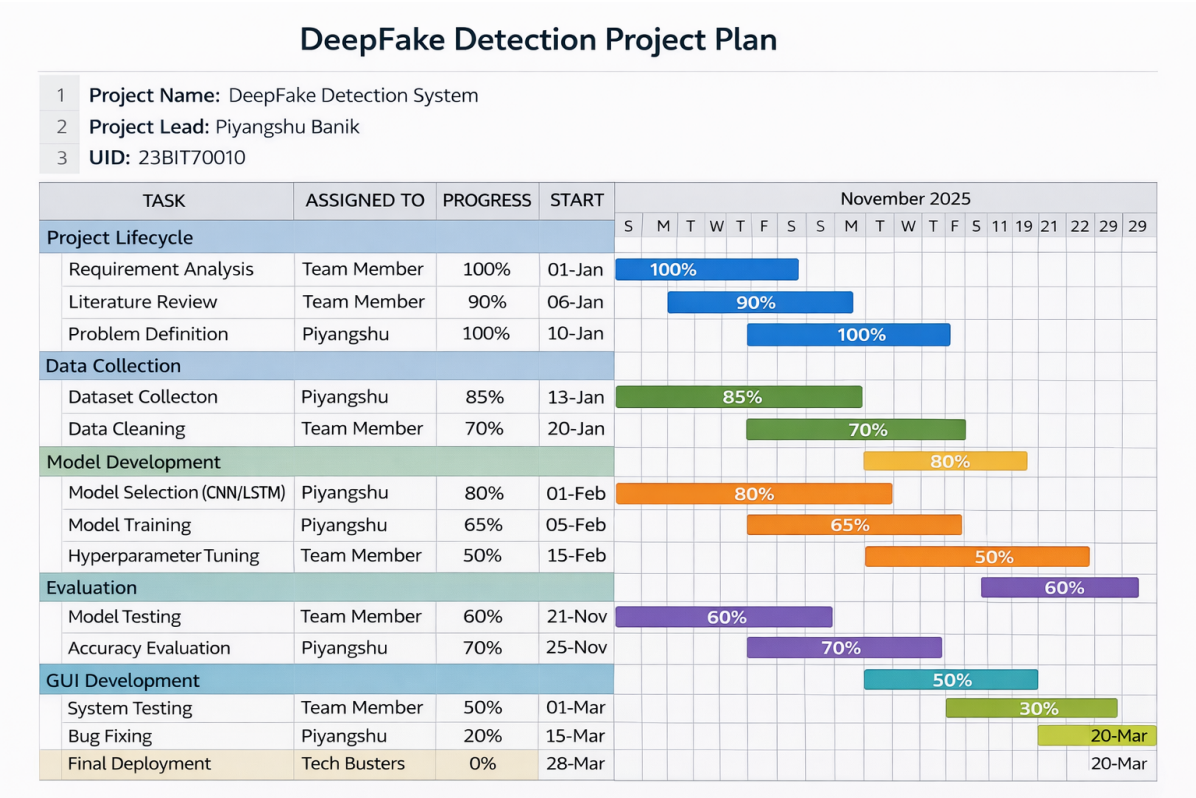

Figure 1: Gantt Chart for DeepFake Detection Project Timeline

Conclusion

The Gantt chart helps in effective project planning and management by providing a clear visualization of tasks and timelines. It enables better understanding of task dependencies, scheduling, and progress tracking.

In this experiment, the Gantt chart successfully represents the development process of the DeepFake Detection System, ensuring systematic planning, efficient resource utilization, and timely execution of all project phases.

Learning Outcomes

- Understanding project planning techniques
- Learning Gantt chart creation
- Managing AI/ML project timelines
- Identifying task dependencies

Evaluation

Criteria	**Marks**	**Obtained**
File Marks	8	
Viva Marks	10	
Conduction Marks	12	
Total	**30**	

Faculty Signature: